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Prediction of diffusional conductance in extracted pore network models using convolutional neural networks

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ABSTRACT

Pore network modeling (PNM) based on networks extracted from tomograms is a well-established tool for simulating pore-scale transport behavior in porous media. A key element of this approach is the accurate determination of pore-to-pore conductance values, which is a complex task that greatly affects the accuracy of flow and diffusive mass transport studies. Classic methods of conductance estimation based on analytical solutions and shape factors only apply to simple pore geometries, whereas real porous media contain irregular-shaped pores. Although direct numerical simulations (DNS) can accurately estimate conductance considering pores' real morphology, it has a high computational cost that becomes infeasible for large tomograms. The present work remedies this problem using a deep learning (DL) approach, with a specific focus on diffusional transport which has received less attention than hydraulic conductance. A convolutional neural network (CNN) model was trained to estimate diffusive conductance of PNM elements from volumetric images of porous media. The developed framework estimates the diffusive conductance by analyzing individual pore-to-pore 3D images isolated from the tomogram to fully capture the topology and shapes. A key outcome of the present work is that only images of the pore regions are used as input data, avoiding excessive preprocessing time for data preparation. The results of the diffusive conductance prediction show good agreement with the test data obtained by DNS method, with 0.94 R^2 prediction accuracy and a speedup of 500x in prediction runtime.

Authorship contribution statement

Niloo Misaghian: Developed 3D deep learning framework, trained and tested the deep learning model, created modules in PoreSpy, drafted the manuscript, Mehrez Agnaou: Developed 3D deep learning framework, trained and tested the deep learning model, Amin Sadeghi: Aided direct numerical simulation, Hamed Fathiannasab: Aided data analysis, manuscript revisions, Isma Hadji: Aided development of 3D Resnet deep learning model, Edward Roberts: Designed the study, and aided interpretation of results, Jeff Gostick: Designed the study, and aided interpretation of results.

1. Introduction

Diffusive transport occurs in a variety of porous media applications such as geological storage of radioactive nuclear waste (Boult et al., 1998; Xiong and Jivkov, 2015), catalyst supports (Ye et al., 2019), and electrodes (Mukherjee et al., 2011; Qu, 2014). The use of pore-scale modeling and simulation techniques to understand the structure-performance relationship of porous materials is common in engineering design, such as optimizing performance of porous components, distribution of catalyst materials, morphology control of undesired deposition (Wang et al., 2014), and others (Andrade et al., 1997; Sadeghi et al., 2017). The two main approaches for pore-scale simulations of transport in porous materials are direct numerical simulations (DNS), and pore network modeling (PNM) (Dong and Blunt, 2009). DNS methods such as the lattice Boltzmann method (LBM) and the finite element method (FEM) have the advantage of probing detailed geometrical features up to the resolution limit of the available image (Borujeni et al., 2013). Their accurate predictions come at a high computational cost that prohibits application in more complex problems such as multiphysics, coupled nonlinear and transient transport

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problems, and large samples where there are billions of unknowns (Blunt et al., 2013). This is in addition to challenges related to building accurate FEM computational grids. On the other hand, PNM is a viable alternative for these large and complex problems due to its vastly lower computational cost (Aghighi and Gostick, 2017; Agnaou et al., 2020; Bonilla and Bhatia, 2012; Fazeli et al., 2016; Sadeghi et al., 2017; Xiong et al., 2015). In PNM the porous medium is abstracted as a connected network where wider void regions are defined as pores (nodes) that are connected by narrower constrictions referred to as throats (edges). This simplification reduces the numbers of variables in the discrete transport equations by orders of magnitude and hence reduces the simulation time concomitantly (Blunt et al., 2013; Sholokhova et al., 2009; Xiong et al., 2016). However, the necessary simplifications to obtain the speed-up decrease the modeling fidelity (Helland et al., 2008). The accuracy of PNM predictions depends on the precise description of the pore space geometry, which is used in the computation of transport conductance values (Baychev et al., 2019). A connected pore-throat-pore element in the PNM is defined as a conduit and the estimation of conduit's conductance for mass transport has been an active research area since the PNM technique was developed (Dehghanpour et al., 2011; Patzek and Kristensen, 2001; Patzek and Silin, 2001; Rong et al., 2020; Sholokhova et al., 2009; Xie et al., 2017). The conduit conductance for a simple cylindrical geometry can be obtained from an analytical solution of the governing transport law (i.e. Fick's law); however, for real pore morphologies the cylindrical approximation loses accuracy. Hence, many attempts have been made to adjust the analytical solution by adding correction factors or shape factors to account for the complex geometry encountered in extracted networks (Dadvar and Sahimi, 2003; Helland et al., 2008; Mortensen et al., 2005; Patzek, 2001; Patzek and Silin, 2001). Most of these efforts have focused on correcting for the deviation of the cross-section from circularity, but they neglect the impact of the diverging and converging nature of the cross-section along its axis. Changes in the constriction have particularly important effects on the diffusion transport conductance (Patzek and Kristensen, 2001) which is the focus of the present work. An alternative to analytical methods based on shape factor is to use DNS to estimate the diffusive conductance of each individual pore network element (Sholokhova et al., 2009; Xie et al., 2017). In this approach, the transport conductance is back-calculated from the DNS results without any arbitrary selection of the pore size and shape information, since the full impact of the conduit geometry is captured in the DNS result. Although this increases the PNM accuracy, it is computationally expensive to conduct this analysis on every tomogram for which a pore network is extracted. A promising alternative, explored herein, is to employ machine learning (ML) algorithms to estimate the conduit's conductance with a relatively low computational cost compared to the DNS approach.

Image-based deep learning (DL) methods have proven to be a powerful tool in classification problems such as cancer detection and disease diagnosis (Hu et al., 2018), age prediction (Chen et al., 2020) where a property of interest is predicted based on only image datasets. They can also be used for regression problems, and in the porous media literature have been used to predict properties such as porosity, permeability, and pores size distribution (Alqahtani et al., 2018; Cang et al., 2018; Javier E. Santos et al., 2020; Sudakov et al., 2019; Wu et al., 2018; Yang et al., 2018). More complex physics-based information has also been obtained. For instance Santos et al. who used CNNs to estimate the gas density profiles in nanopores of shale samples (Santos et al., 2020). The advantage of using CNNs over using regular neural network models to predict effective properties of porous media was established in a previous work (Srisutthiyakorn*, 2016), which attributed the success to its convolutional architecture which explicitly assume that spatially close inputs are semantically related. This assumption is an advantage in image data where the features are obviously spatially related. In the present study, a CNN framework was developed to extract and learn the spatial features of random irregular-shaped conduit images and predict diffusive conductance values.

The prospect of using ML to estimate the conductance of PNM elements has recently attracted attention (Rabbani and Babaei, 2019). The general idea is that sub-images of each conduit can be extracted from the full tomogram, by performing a watershed segmentation to identify pore regions, then each pair of pores are analyzed to predict their conductance. Both Miao et al. (2017) and Rabbani et al. (Rabbani and Babaei, 2019) trained a model to predict hydraulic conductance, such that Darcy flow could be simulated in the resultant PNM. Because they both focused on hydraulic conductance, the physics were dominated by the throat constriction, to the extent that in both cases the ground truth or training data was generated by using LBM to simulate flow through the throat cross-section only, neglecting the effect of the larger pore bodies. Both groups then estimated some relevant geometrical parameters about the throat size and/or shape (convex hull convexity and circularity (Miao et al., 2017), and diameter of an inscribed sphere (Rabbani and Babaei, 2019)) for training the ML model. In other words, the ML model was trained to predict the given conductance values obtained by DNS simulations on the throat cross-section, from knowledge of only a few geometrical properties of the throat. There are two key limitations to this approach. First, viscous flow is highly dependent on the tightest throat constriction (*i.e.*, $\sim R^4$ where R is the radius of the conduit), whereas diffusion is less sensitive to this (*i.e.*, $\sim R^2$) and thus more impacted by the size and shape of the entire conduit. Therefore, obtaining training data based only on transport through the throat constriction is insufficient. Second, since the entire conduit is of interest, it becomes much less obvious which specific geometrical properties should be extracted and provided to the ML model as input. Moreover, this would require extracting multiple metrics using tools that are themselves computationally demanding (*i.e.*, fast marching method, skeletonization, local thickness, *etc.*). It is therefore an open question how to best leverage ML to estimate diffusive conductance values for use in PNMs.

The proposed workflow in this study is based on training a CNN model to predict the diffusive conductance of pore network elements based on only images of neighboring pore region pairs. The presented study estimates the diffusive mass transport properties and tackles defining challenge of previous ML-based methods by using a non-preprocessed image-based framework. This framework trains a DL model simply by providing a voxel image of two connected pore regions, with each region labeled. This is the smallest possible amount of preprocessing that can be done on input data. The proposed framework substantially decreases the preprocessing time and resolves the challenge of finding conduits features. The DL model can be used to predict and assign physics properties of conduits (diffusive conductance) in pore network models of porous media. This approach can be generalized in the future to predict other properties. This manuscript is organized as follows. First, the theoretical background and the methodology of the framework are explained. Then, the DL data and trained model are described. Finally, the pore network simulation results and their comparison with currently available PNM approaches (Helland et al., 2008; Patzek and Silin, 2001; Xiong and Jivkov, 2015) and DNS methods are presented and discussed.

2. Theoretical background

2.1. PNM formulation and conduit conductance

Diffusion of a species in a conduit is described by Fick's first law:

$$J_n = -\left(\frac{D_n A}{L}\right)\Delta C_n, \tag{1}$$

where J_n and C_n are the diffusive mass flux of species n, and its concentration, respectively. In Eq. (1), D_n is the diffusion coefficient of species n in the solvent (note that this is the diffusion coefficient in an open space, not an effective diffusivity), and A and L are the area and

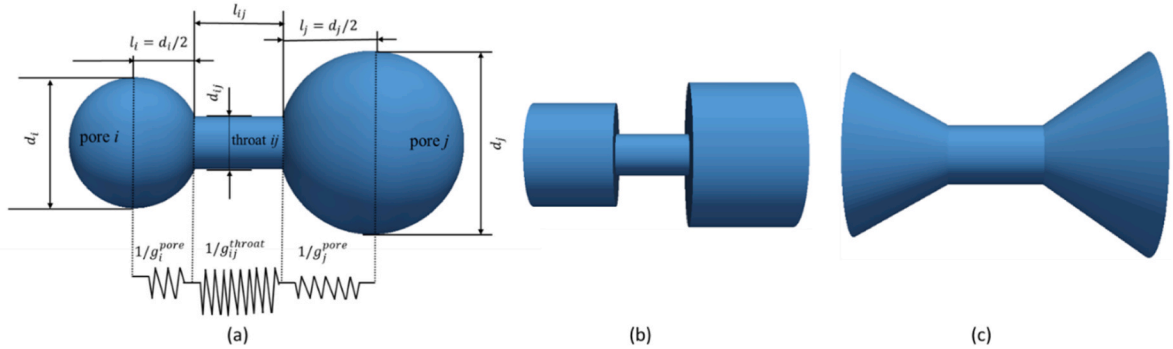


Fig. 1. A schematic of different geometries for a pore-throat-pore element as one conduit. a) Simple sphere and cylinder geometry. The conduit elongates from the centroid of one pore and includes their connecting throat and half of two pores. The corresponding conductance of the conduit is calculated using the linear resistor theory for resistors connected in series on its elements (half pore-throat-half pore) from Eq (4) [43]. b) Cylindrical tubes for pores and throats, referred as geometry-based method. c) An example of a conical frustum and cylinder. The corresponding conductance of the conduit is calculated using the series of resistors model with a correction, referred as shape factor based method.

length of the conduit. If the conduit is a straight cylinder, A and L are well defined; however, if the conduit's shape is irregular and cross-section area along its length varies, these become *effective* values A_e and L_e, respectively. Specifically, when the conduit connecting pores i and j is irregular-shaped, Eq. (1) should be amended as follows (Xiong and Jivkov, 2015):

$$J_e = -\frac{D_n A_e \Delta C_{ij,n}}{L_e} = -G_{ij}^{diff} \Delta C_{ij,n} \tag{2}$$

where G_{ij}^{diff} is the overall diffusive conductance of the conduit connecting pores i and j. Applying this expression to each conduit in the network and enforcing mass conservation at every pore i yields the following system of equations, assuming steady-state:

$$\sum_{j=1}^{N} G_{ij}^{diff} \left(c_{i,n} - c_{j,n}\right) = 0, i = 1, 2, \ldots, k \tag{3}$$

where N is the total number of pores connected to pore i and k the total number of pores in the network. Modeling Fickian diffusion in a pore network requires solving this system of linear equations (Eq. (3)) assuming relevant boundary conditions are applied. This results in concentration values in each pore of the network. It is apparent that G_{ij}^{diff} values directly affect the accuracy of the simulation, so a network extraction tool must be able to reliably determine these values.

A conduit is described with the combination of ½ of pore i, ½ of pore j, and the throat connecting them. Given that a conduit has three elements in series, G_{ij}^{diff} is the cumulative conductance of each element which can be defined based on the linear resistor theory for resistors connected in series by:

$$G_{ij}^{diff} = \left(\frac{1}{g_i^{diff}} + \frac{1}{g_{ij}^{diff}} + \frac{1}{g_j^{diff}}\right)^{-1} \tag{4}$$

where $g_{(i \mid ij \mid j)}^{diff}$ is the conductance of each element (half pore i/throat ij/half pore j) in the conduit.

Fig. 1 shows a pore-throat-pore conduit and different regular shapes that can be assigned to pores. In the conventional approach for calculating conductance, each element in the conduit is usually assumed to be a cylindrical tube (Fig. 1b). This arrangement, referred to herein as geometry-based conductance model (Gostick et al., 2007), yields:

$$g_i^{diff} = \frac{\pi D}{4}\left(\frac{d_i^4}{l_i}\right) \tag{5}$$

where d_i and l_i are the diameter and length of the elements. Equation (5) applies to pores i and j and the throat ij. The length l of a pore is its radius as shown on Fig. 1a. Inserting equation (5) into Eq. (4) yields G_{ij}^{diff} values, which can then be used in Eq. (3). To determine g_i^{diff} from tomograms, it is necessary to extract meaningful values of d_i and l_i of the pore and throat elements. This is a challenging task in complex 3D media where the void space is highly irregular in shape, and the pores and throats cannot be completely distinguished.

As explained in the Introduction, different studies have proposed shape factor correction models to salvage the use of the geometrical model for irregular geometries. A more accurate shape factor correction can be obtained by accounting for a variable cross-section in a pore/throat (Akbari et al., 2011). Assuming a diverging conduit of connected conical frustums and cylinder, as shown in Fig. 1c, the shape factor ($S_{(i \mid ij \mid j)}^{diff}$) can be calculated as:

$$S_{(i \mid ij \mid j)}^{diff} = 1 \Big/ \int \frac{dx}{A_{(i \mid ij \mid j)}(x)}\, S_i = \frac{d_i d_t \pi}{4 l_i},\ S_j = \frac{d_j d_t \pi}{4 l_j}\ S_t = \frac{\pi d_t^2}{4 l_t} \tag{6}$$

where the shape factors are multiplied by geometry-based diffusive conductance of each element as mentioned above:

$$g_{i_shape_factor}^{diff} = S_i^{diff}\, g_i^{diff} \tag{7}$$

In this study, shape factor based conduit conductance values are calculated for comparisons, using Eq. (7). Despite its advantages over geometry-based method (Eq. (5)), the diffusive conductance calculated based on the shape factor method still results in erroneous predictions of overall effective diffusivity of the network compared to a DNS method using the image.

2.2. Image-based framework

Instead of attempting to extract accurate values for quantities describing the geometry, it is possible to find G_{ij}^{diff} using DNS instead. This can be achieved by performing DNS of diffusion on each sub-image representing two neighboring pores and calculating G_{ij}^{diff} using Eq. (2). This approach would provide increased accuracy, but is computationally expensive given that a typical tomogram may require thousands of such mini-simulations. So, although this has been demonstrated to be feasible (Sholokhova et al., 2009) it is not generally practiced.

The objective of the proposed workflow is to overcome the issues of estimating the diffusive conductance of PNM elements accurately and rapidly. For this purpose, a DL model was trained to predict diffusive conductance of pore network elements with only images of the pore-to-pore regions as the input data. The bounding box containing a pair of pore-to-pore region is referred as region of interest (ROI). The input

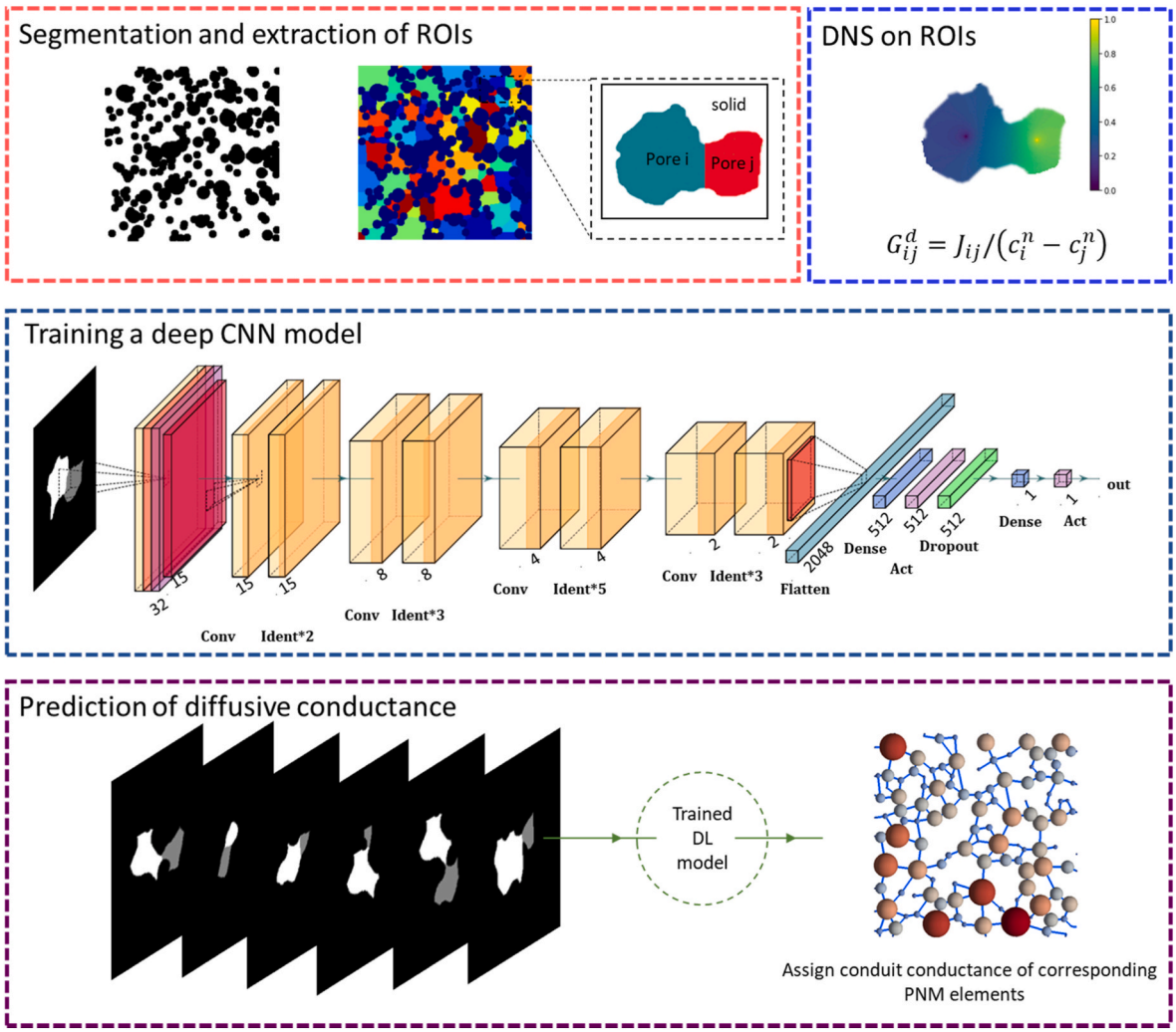


Fig. 2. *A flow chart of the proposed algorithm. It starts by applying an improved watershed segmentation (SNOW algorithm (Gostick, 2017)) on binary images of porous media (top left). For each region of interest (ROI) in the segmented image (real shapes of pore-throat-pore element) the diffusive transport is simulated using DNS to find its diffusive conductance value (top right). Labeled images of these regions are passed to a DL model (Resnet 50 (He et al., 2016)) to extract their features and train on existing images with their DNS conductance as the ground truth data (middle). The model performance is then assessed with a non-biased test dataset (bottom).*

image data included random irregular-shaped images of the ROIs extracted from segmented regions of randomly generated porous media images. The randomness in the shapes of ROIs is because of segmentation on randomly shaped void space of each porous medium. The details of generated images are given in section 2.3. The ground truth data was calculated by implementing diffusion DNS on the pore-to-pore images. A CNN model was then built and trained on the image dataset. The trained model was used to predict the diffusive conductance of unseen image datasets. The model performance was assessed based on prediction accuracy and runtime. Note that the framework was initially implemented

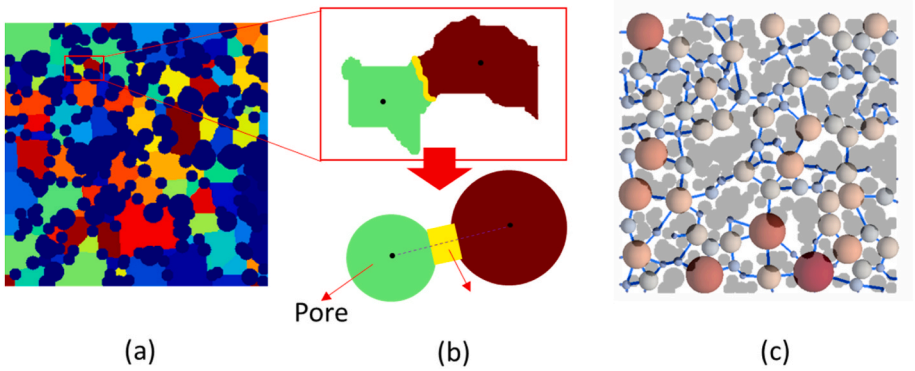


Fig. 3. Illustration of the watershed segmented image and an ROI for a 2D randomly generated porous medium. a) Segmented regions based on watershed method. b) Two neighboring pore regions segmented image (top) and their corresponding pore and throat model in a PNM (bottom). c) Pore network connections in the void space.

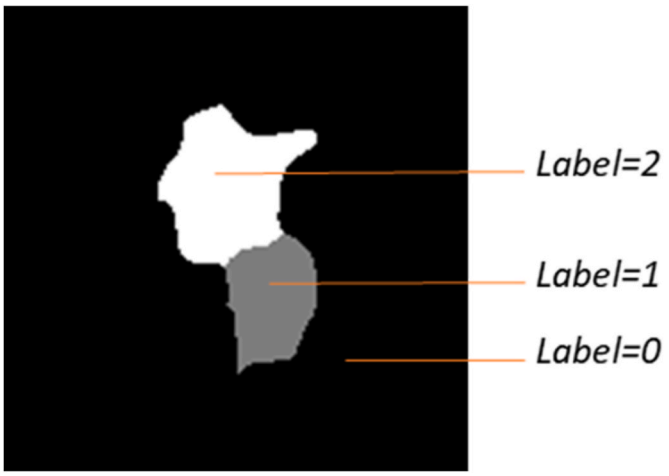


Fig. 4. A sample input image for the ML algorithm. It includes two connected pore regions extracted from the whole medium segmented void space. To remove the effect of label values on the algorithm, solid-phase along with the regions other than ROI are labeled as 0. Two connected pore regions in the ROI are labeled as 1, and 2, respectively.

for both two-dimensional (2D) and three-dimensional (3D) images. The final model presented in this study was developed based on 3D images. However for visualization purposes, the 2D samples are shown in the framework and in the figures in following sections. The workflow in this study is illustrated in Fig. 2. Details of each step in the workflow are presented in the next sections.

2.3. Preparation of the training dataset and ground truth labels

To cover a large variety of geometrical shapes, topologies, and geometrical complexities in the training data, synthetic images of porous media were generated. For training a DL model, synthetic image generation can provide a large number of samples, which is difficult to obtain from a limited number of real porous media images. The 3D images of random porous media were generated using PoreSpy's random polydisperse_spheres generator (Gostick et al., 2019). PoreSpy is an open-source collection of image analysis tools used to extract information from 3D images of porous materials. PoreSpy's polydisperse_spheres generator module creates an image of randomly placed, overlapping spheres with a distribution of radii, and the shape and overall distribution of the void/solid phase are varied. The distribution of overlapping spheres' radii was defined using Scipy's (Virtanen et al., 2020) statistical normal distribution function. To include varieties of realizations close to real geometries, a wide range of porosities (0.35–0.8) were used to generate the random images.

The generated images are sets of 3D binary images from which a pore network can be extracted. For this study, sub-images (ROIs) of the pore-to-pore regions of pore network elements were required. These ROIs can be captured from watershed segmentation of images. Image segmentation is the first step towards pore network extraction where the binary image of the porous medium is divided into segmented pore regions. In this work, a custom marker-based watershed segmentation method (SNOW method (Gostick, 2017)) in PoreSpy was used to segment the porous media images (Fig. 3a). Each region in the segmented image corresponds to a pore, and the constriction to its neighboring region corresponds to a throat in the pore network model (Fig. 3b bottom). Therefore, the ROI for this study is two neighboring areas in a segmented image of a porous medium that corresponds to a pore-throat-pore element (conduit) in the pore network model (Fig. 3b top).

The corresponding pore network of the segmented image is illustrated in (Fig. 3c). The framework developed in this study only requires the information of the image before network extraction (*i.e.*, Fig. 3a). Therefore, the input images of the framework are ROIs of the segmented samples. For each ROI, two connected pore areas were labeled 1 and 2 and the rest of the segmented image was labeled 0 and treated as non-conducting phase. Flipping the labeling of the regions in every ROI did not notably alter the predicted conductance of the conduit.

This label-based description of the pairs was found to work as well or better than other different preprocessing steps, such as distance transform, in terms of DL model performance. The labeled image of pore-throat-pore captures the shapes and irregularities of pore regions and the location of the throat as shown in Fig. 4. Therefore, the converging and diverging parts of the conduit shape are also well captured. The labeled ROI images were defined as input data for the DL workflow. The resolution of all images was 1 $pixel^{-1}$. Therefore, the framework is independent of the resolution of the images, as the diffusive conductance values are calculated in terms of voxels with the dimension of 1/pixel in 2D and 1/voxel in 3D.

To improve representativity of the images dataset and remove any bias related to image generation, augmentation methods including rotations and flips were used. (Hu et al., 2018; Smith and Topin, 2019).

These ROI images were also computational domains for calculating the ground truth data. A diffusion problem at the pore-scale over the

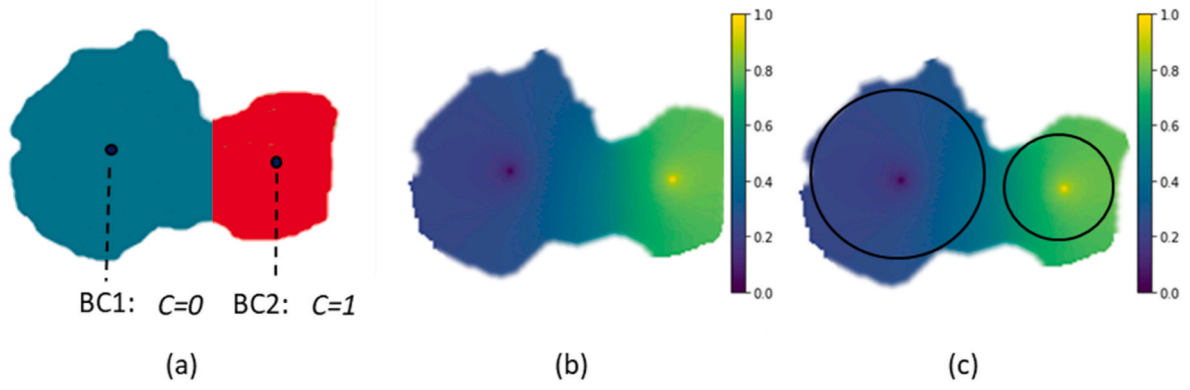


Fig. 5. Steps towards the calculation of 2D conduits diffusive conductance using DNS. a) After meshing the conduit using a square (hexahedral for 3D conduits) grid, Dirichlet concentration boundary conditions were imposed at the pores centroid. b) Concentration color map resulting from the solution of the diffusion problem. c) An illustration of the portions of the 2 regions lying within the corresponding inscribed circle (sphere for a 3D conduit). Surface (volume for 3D conduits) average concentration over the inscribed circles (spheres in 3D) are used for the calculation of conduit's effective properties (i.e., the diffusive conductance). An inscribed circle is the largest circle that can fit into a pore region (Gostick, 2017) while having the pore's centroid as a center. Inscribed circles are obtained during the network extraction using PoreSpy (Gostick et al., 2019). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

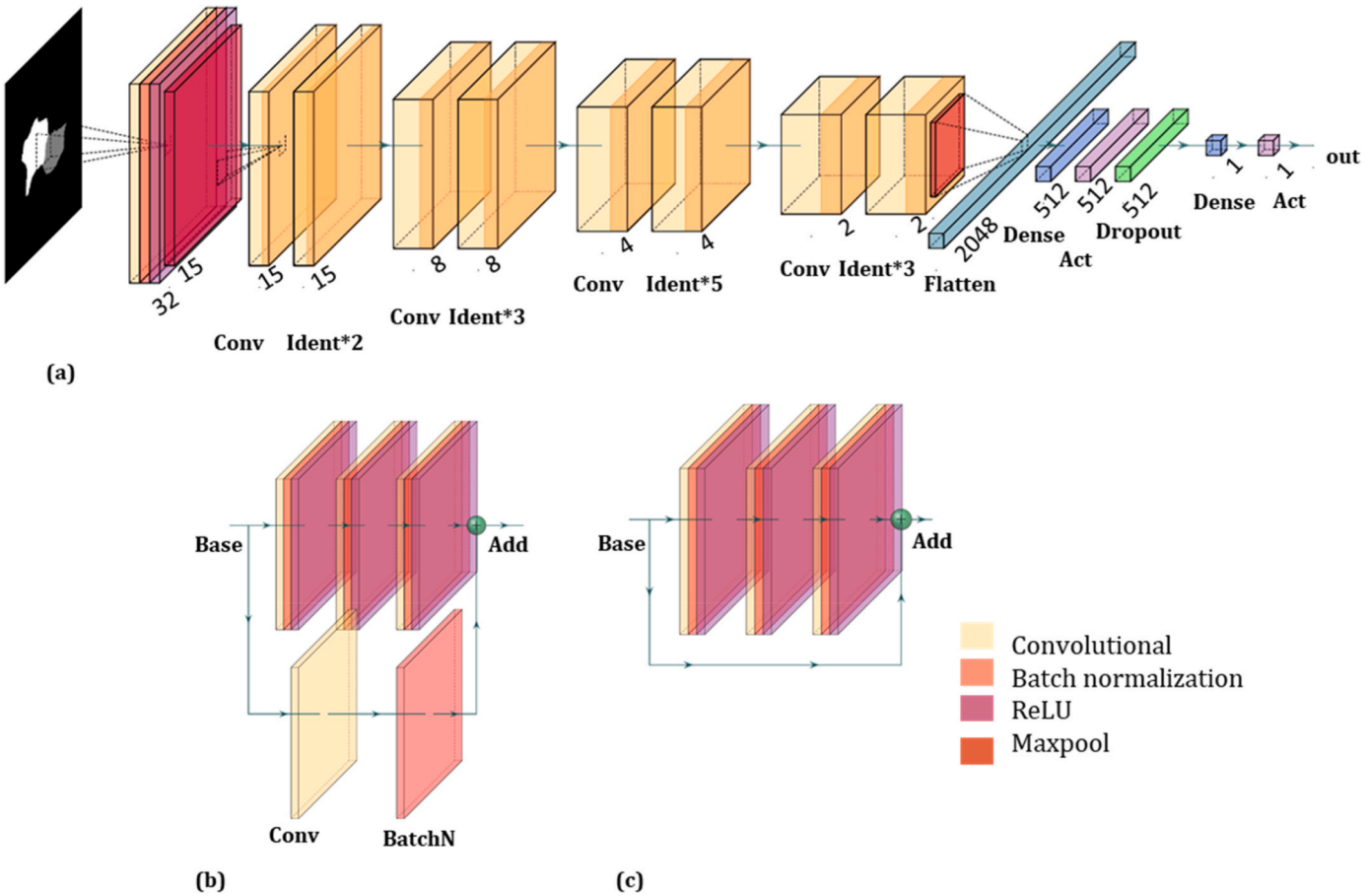


Fig. 6. The architecture of the CNN model used in this study. a) Resnet model. b) A convolutional block (conv blocks are repeatedly used in the Resnet model with different kernel sizes. c) An identity block (identity blocks are repeatedly used in the Resnet model), the bypass identity connection allows the input to pass through the cell without modifications.

conduit was defined and solved using DNS (finite difference method) as described in Fig. 5. A square (or cubic for 3D conduits) grid was used to discretize the computational geometry. For each ROI, Dirichlet concentration boundary conditions were imposed on the centroids of the two pore regions. After solving the resulting linear system of equations, the diffusive conductance was computed using in Eq. (2) (Xiong and Jivkov, 2015) with concentrations averaged over the portions of the domain within the two inscribed spheres (or circles for 2D images) (see Fig. 5 (c)). In the rest of this work, the DNS based conductance values are referred as the 'ground truth'.

2.4. Deep learning model

The building blocks of a simple CNN are stacks of convolutional layers (CL) followed by maxpooling layers and a fully connected layer as the last layer to produce the output (Hope et al., 2017). A CL extracts the patterns and features of an image by applying filters to the image (Ramsundar and Zadeh, 2018).

The CNN architecture in this study was developed based on a residual network model Resnet50 (He et al., 2016) that is advantageous in addressing "vanishing gradient problem" in conventional deep networks (Pascanu et al., 2013). The vanishing gradient problem is addressed by adding bypass connections where parts of the transformed image can pass through deeper layers without losing their strength (Fig. 6 c). The CNN structure in this study is illustrated in Fig. 6. The developed model is a 3D Resnet50 that takes in 3D voxelated images of ROIs as input. However, for illustration purpose a 2D image is shown in Fig. 6. The model starts with a 3D CL with a cubic kernel that sweeps on a padded input image to extract features directly from images. The CL is followed by a batch normalization layer where the convoluted input image is normalized based on the samples in the same mini-batch. The resulting image is passed to an activation layer with an activation function of rectified linear unit (ReLU) $f(x) = \max(0, x)$. Downsampling is then applied to the image using a maxpooling layer. The remaining building blocks of the network are convolutional and identity blocks. Convolutional and identity blocks include a pattern of convolution, batch normalization, and activation layer with bypasses explained above (Fig. 6 b,c) (He et al., 2016). After a series of convolutional blocks and identity blocks, the last block of the network starts with a pooling layer. The output of this layer is then flattened to be fed into a dense layer. A dropout layer at the end was used to prevent the overfitting problem and increase computational efficiency.

3. Results and discussion

3.1. Training process

The workflow for training the DL model was developed in Python ("Python," n.d.) using TensorFlow (Abadi et al., 2015). From a dataset containing 24,465 images of pore pairs, the training and validation sets were randomly selected in a way that 80% of the images were used as training data and 10% of the images were used as validation dataset. To assess the prediction performance of the model for unseen data, the remaining 10% of the images were used. The target data was normalized with min-max scaling between $[0, 1]$ based on training dataset.

The loss function for this regression problem was defined as the MSE

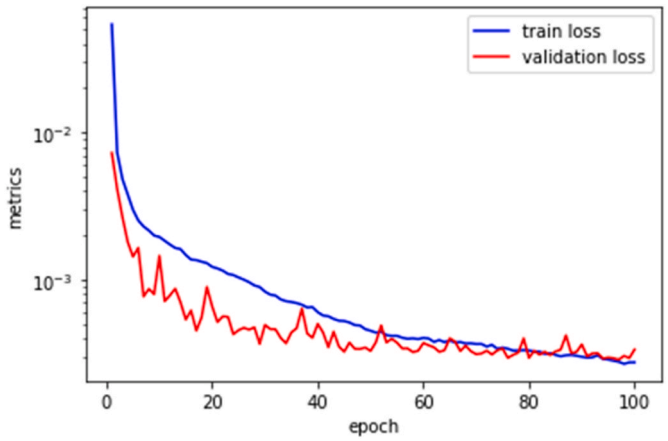


Fig. 7. Loss function values for the training and validation dataset for 100 epochs (semi-log scale).

between the predicted and target values of diffusive conductance. This function was selected among other functions including Huber and root mean squared error (RMSE). The objective of the optimization process in CNN learning is to minimize this loss function as described in Eq. (8).

$$\text{Loss} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - Y_i^{'}\right)^2 \tag{8}$$

where n is the number of data (labeled images of two regions), Y is the vector of target values and $Y^{'}$ is the vector of predicted values of diffusive conductance.

The model was fitted in the training process by minimizing the MSE loss function using Adam Optimizer with a learning rate of 10^{-4}. The prediction accuracy was defined as R^2R^2 (coefficient of determination) accuracy which indicates the proportion of variance of target values that were defined by variables in the model as following: [47]:

$$R^2\left(Y, \acute{Y}\right) = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - Y_i^{'}\right)^2}{\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2} \tag{9}$$

where $\overline{Y}$ is the average of target values calculated as. $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$.

It must be emphasized that training the model requires setting the hyper-parameters of the DL algorithm and adapting its architecture to the task. This was done by performing the training several times, adjusting one parameter at a time, and assessing the performance and accuracy of the model to find the best values for each parameter. Using this process, a batch size of 16 was and a dropout probability of $P = 0.5$ were selected as the best values in terms of memory and accuracy efficiency. To increase the memory efficiency of the workflow all the input and output files were saved in Hierarchical Data Format (HDF) files.

The framework was developed on a machine with 2.10 GHz Intel Xeon CPU. Model training was implemented on a GPU architecture using Nvidia Quadro RTX 8000 graphics card. Each epoch of training took approximately 8.2 min to run. Training the model took about 13.7 h. After training, the model predicted conductance of 2447 images of 3D conduit test data in 9.32 s with R^2 accuracy of 0.97. That is approximately 0.0038 s to predict each target label. In contrast, predicting the diffusive conductance on those conduit images using DNS method takes between 2 and 10 s (average 3.5) depending on the size of the image for the test dataset. DNS based runtime increases as the size of each pair of image becomes larger. This performance difference can become quite substantial when networks are extracted from images containing 1000's of conduits. The trained DL model takes in tensors of large number of images and can predict the results within seconds. As the number of conduit images increases, the computational cost of DL-based prediction does not increase significantly. On the other hand, the DNS-based prediction of larger number of images becomes more time consuming. A detailed comparison of these two PNM methods is shown in the next sections.

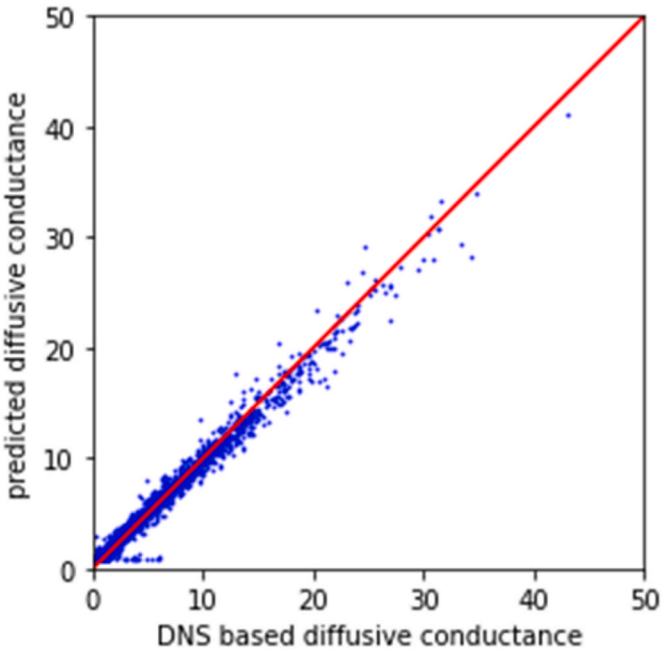


Fig. 8. Predicted diffusive conductance vs DNS based diffusive values (ground truth) for 2447 test images tested on the fitted model.

3.1.1. Training and validation loss

The loss values for training and validation datasets in each epoch of the training are shown in Fig. 7. The decreasing trend in training and validation errors indicates an improvement in the CNN model capability to estimate target values as more epochs are completed. The training and validation errors reached a plateau of approximately $MSE = 2.7563 \times 10^{-4}$ and $MSE = 3.3711 \times 10^{-4}$ at 100 epochs, respectively. The trend indicates that overfitting or underfitting did not occur when the model was stopped at 100 epochs. However, we observed that before epoch 85, the validation loss is lower than the training loss. There are two main reasons that can lead to this trend in early stages of the training until reaching an optimum point of a good fit (epoch 100). First, because there is a competition between optimization and generalization, the optimization tends to update the model to best perform on the training data whereas the generalization tends to have a model that performs well on unseen data (test or validation data). At the early stages of the training, these two concepts are correlated (Chollet, 2021) and the model can be underfit, meaning that there is still room for improvement of the model and the network has not captured all important patterns in the training data. At the later stages of the training, the generalization stops improving and the model starts to overfit meaning that the model is learning the patterns that are more specific to the training data. A good fit will be where both optimization and generalization are in balance and the training and testing loss are low but not very much different in terms of order of magnitude. This condition of a good fit was reached at epoch 100. Second, this may be related to the use of dropout layers to prevent overfitting (Chollet, 2021). During the training the dropout layer drops some of output features of the layer so that the optimization focuses on the most important patterns in the training data instead of all patterns, thereby allowing the network to generalize better (perform better on unseen data). As more epochs are processed, the trained network improves and both training and validation loss become closer until they reach a plateau near epoch 100.

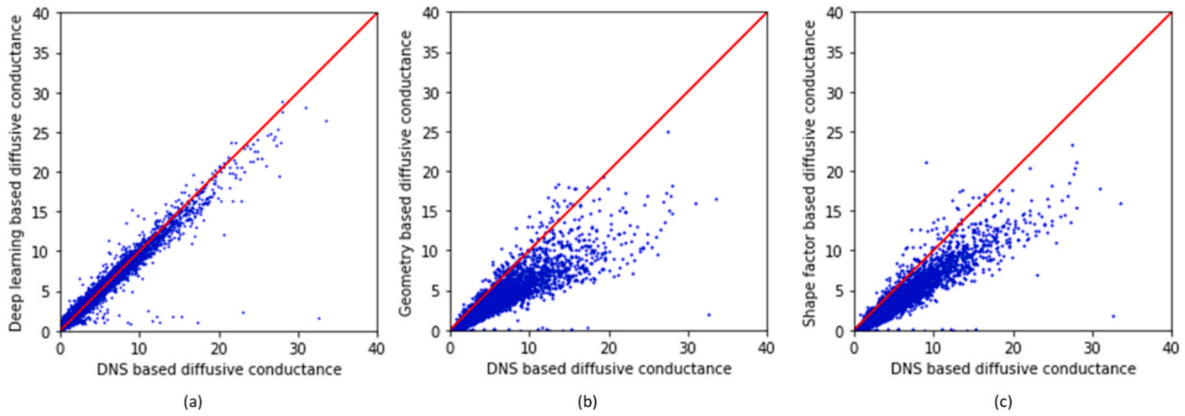


Fig. 9. *Predicted diffusive conductance obtained by different methods vs DNS based values for 6163 test images. a) Predicted diffusive conductance using DL model vs ground truth values, R^2 accuracy = 0.94. b) Geometry-based diffusive conductance vs ground truth values, R^2 accuracy = 0.62. c) Shape factor based diffusive conductance vs ground truth values, R^2 accuracy = 0.71.*

3.2. Prediction

3.2.1. Prediction accuracy

The ability of the trained model to predict diffusive conductance of unseen data (vs Fickian diffusion equation solution obtained by DNS method described in Section 2.3) was then assessed. Prediction results for unseen data shown in Fig. 8 indicate a high accuracy of prediction up to $R^2 = 0.97$.

To compare the proposed framework with traditional methods of estimating diffusive conductance, a separate larger dataset of 6163 conduit images were used as the test data. The conduit images were extracted from a new set of 14 randomly generated images. The DL-based prediction results for these unseen images are shown in Fig. 9.

After extracting the conduit images the DL prediction runtime for 6163 conduits was 13s whereas the DNS prediction was 2.5 h. The advantage of using DL-based framework to predict the values is more noticeable here, as after increasing the number of images from 2447 (shown in Fig. 8) to 6,163 (shown in Fig. 9) the DL-based prediction runtime does not increase significantly (from 9.32s to 13s), which is about 4 s increase while still remaining in the order of seconds. On the other hand, the DNS-based prediction runtime increases significantly (from 1.5 h to 2.5 h), which is about 1 h increase in runtime.

For comparison, also shown in Fig. 9 are the conductance values for the same set of images but calculated with the basic geometrical method and shape factor based method both discussed in section 2.1. The geometry-based conductance values (Eq. (5)) and shape factor based methods (Eq. (7)) are underestimating/overestimating the conduits' conductance resulting in a higher estimation error. The accuracy of these methods to estimate diffusive conductance can be even lower in more irregular-shaped geometries and heterogeneous media. As shown in Fig. 9, geometry-based conductance values have a large deviation from ground truth values resulting in the lowest estimation accuracy. This demonstrates the inaccuracy of conductance values calculated based on simplified geometries. Attempting to correct these geometrically computed values using shape factors improves the overall result marginally, as shown in Fig. 9c. Despite some improvement, the shape factor based method still presents a large deviation from ground truth data as degrees of irregularities in pore shapes increase. The main reason is shape factor methods are only partially successful in that they do not account for the diverging/converging nature of the conduits. Conduit features such as diverging/converging section, shape and more complicated features that cannot be quantified as a factor such as spatial features were automatically extracted in a CNN framework. As shown in Fig. 9a, the trained DL model estimated conductance values for unseen

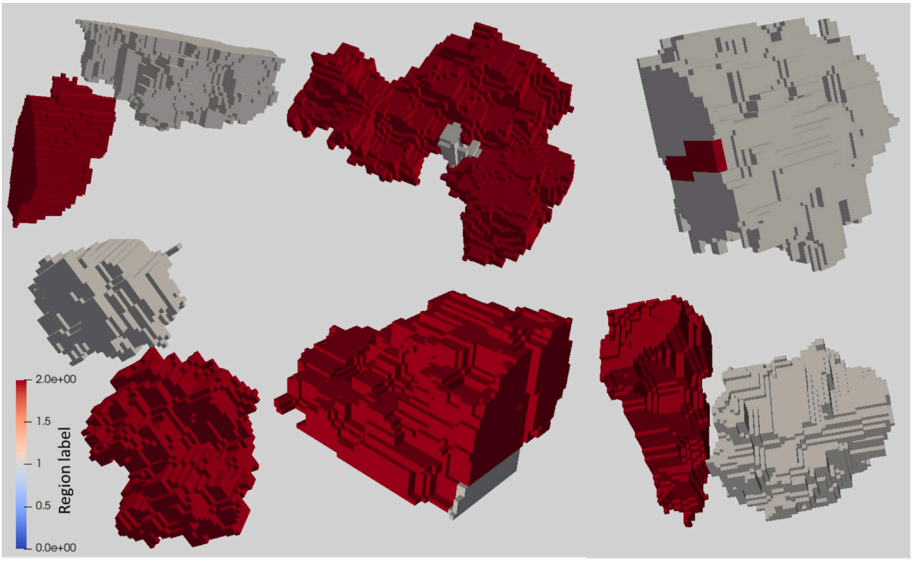


Fig. 10. Examples of ROIs (from 6163 test images) that the trained model predicts their diffusive conductance with the least accuracy. These ROIs correspond to the scatter points in Fig. 9a that are along the bottom close to the horizontal axis.

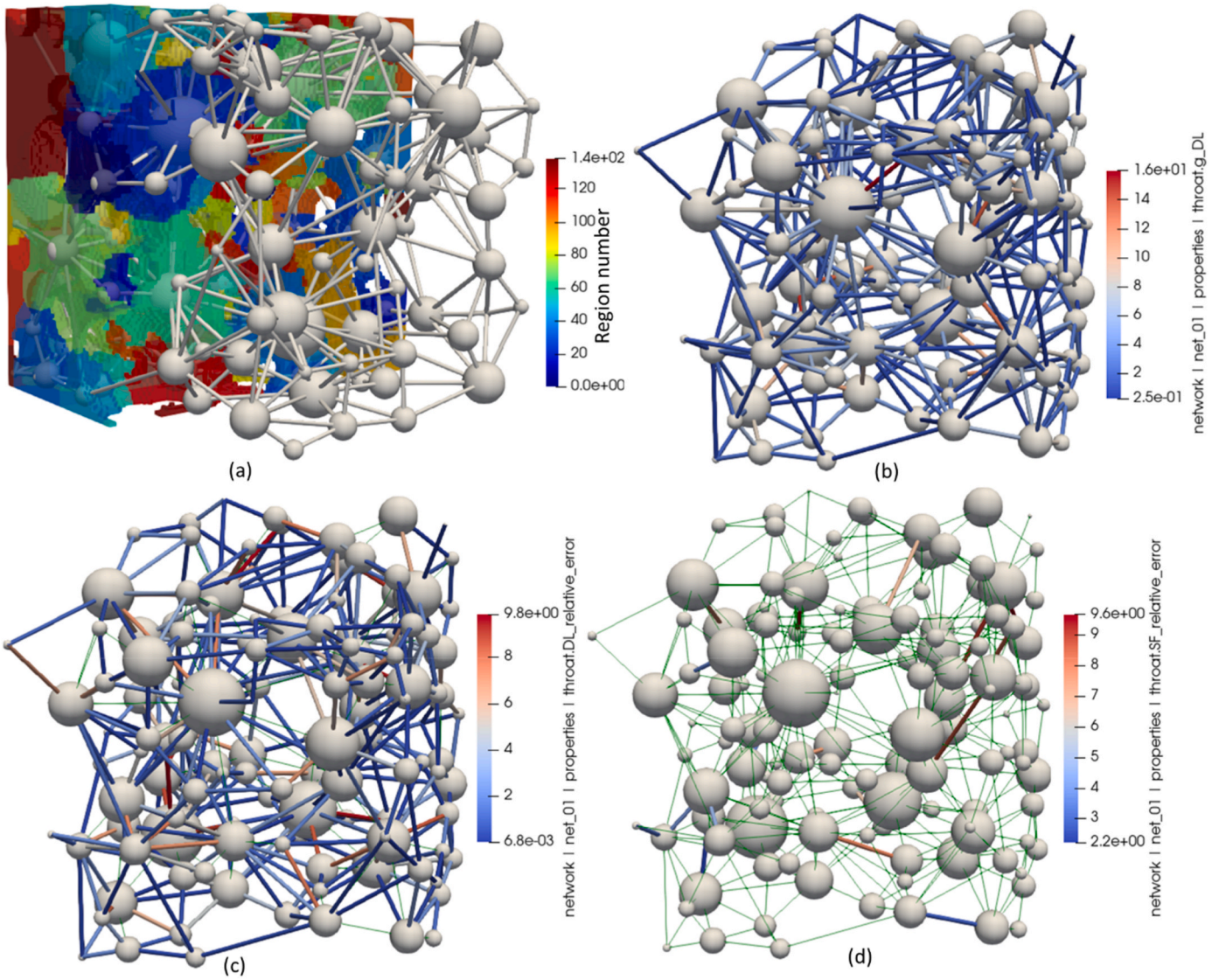


Fig. 11. Pore network visualization of a randomly generated porous sample. a) Part of the segmented image of the porous sample with the extracted pore network of the sample's void space. Each segmented region is colored by its index value. b) The pore network throats are colored by predicted diffusive conductance of conduits using the DL method. c) The pore network throats are colored by the percent error of DL-based conductance method. The throats with higher than 10% error are distinguished with thin green lines. The DL prediction accuracy for diffusive conductance of all conduits in the network was $R^2 = 0.94$. d) The pore network throats are colored by the percent error of shape factor-based conductance method. The throats with higher than 10% error are distinguished with thin green lines. The shape factor prediction accuracy for diffusive conductance of all conduits in the network was $R^2 = 0.65$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

images of conduits with the highest accuracy ($R^2 = 0.94$) in contrast to traditional geometrical and shape factor based methods (Fig. 9). However, there are some points where the predicted conductance is much lower than the DNS value (the points along the bottom close to the horizontal axis). To find the reason for the deviation, the volumetric image of pore-to-pore regions corresponding to those points were investigated. It was found that those regions either contained a connecting throat of one to two voxels or had a slit-like geometry embedded into their large neighboring pore. Examples of such ROIs are shown in Fig. 10. These two types of properties are rare in the segmented image dataset and are not a desired property for PNM in general. Therefore, these images were because of the original image's highly constricted region, and image segmentation method that could be improved to modify the segmentation.

To compare the DL predicted values with shape factor method visually, an example of a randomly generated porous sample with the size of 100^3 voxels is shown in Fig. 11. The extracted network model and part of the segmented image of the porous sample is shown in Fig. 11a. The pores are represented as spheres (shrink to half of their size) and throats are represented as cylinders with a constant thickness of 5 voxels. This scaling of the size was done only for visualization purpose. In other words, the pore network model and its geometry model used for calculating the diffusive conductance includes the real values of pore and throat sizes. Predicted values of conduit conductances from DL model is shown in Fig. 11b. Each throat is colored by the predicted conductance value, which spans on a range of [0.25,16] for this porous sample. The DL prediction accuracy was $R^2 = 0.94$ while the shape factor method's prediction accuracy was $R^2 = 0.65$. Therefore, the DL method predicted the conductance values more accurate than the shape factor method. To compare the predicted diffusive conductances in more details, the percent error of estimated conductance of each individual conduit was calculated using $\textit{percent error} = \left|\frac{\textit{estimated value} - \textit{DNS value}}{\textit{DNS value}}\right| \times 100$. As a visual comparison, the percent errors of DL conductance and shape

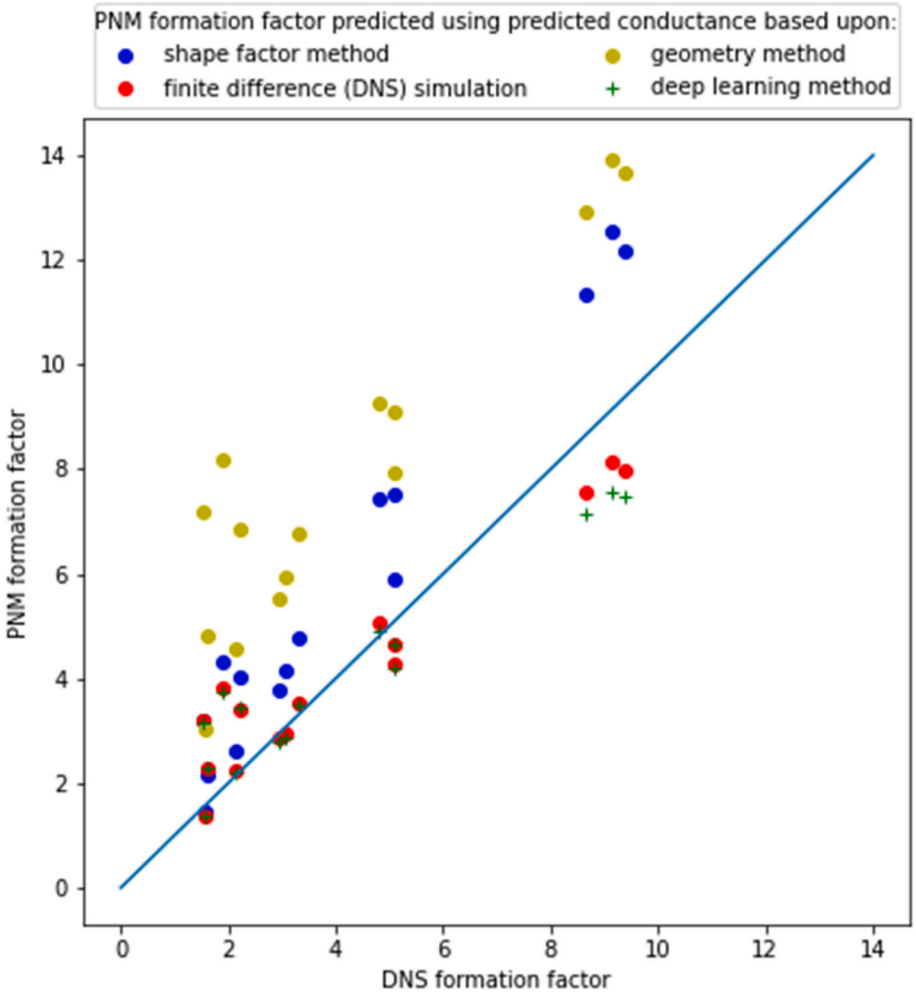


Fig. 12. A plot of formation factor calculated using PNM vs DNS for 15 images. Blue dots data points are the PNM with shape factor based diffusive conductance, red dots data points are the PNM with finite difference-based diffusive conductance, yellow dots data points are the PNM with geometrical diffusive conductance, and green plus data points are the PNM with our DL-based diffusive conductance. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

factor method conductance are shown in Fig. 11c and d. For better distinction of the difference in the errors, only the throats with percent error smaller or equal to 10% were colored in Fig. 11c and d. The throats with percent error higher than 10% were represented by thin green lines in Fig. 11c and d. As shown in Fig. 11c, the minority (15% of all conduits in the network) of the percent error are higher than 10%, whereas in Fig. 11d, the majority (96% of all conduits in the network) of the percent error are higher than 10%. This comparison shows the advantage of DL method over the shape factor method to predict the conductances of conduits. Images shown in Figs. 10 and 11 were visualized in ParaView.

3.3. Deep learning-based PNM comparison with DNS on the entire domain

The accuracy of PNM results using the local diffusive conductance values found from the proposed DL framework was assessed with reference to the results from the DNS method on the entire domain. The formation factor prediction for 15 randomly generated synthetic porous media calculated using PNM with DNS-based conduit conductance, DL-based conduit conductance, PNM with geometry-based conduit conductance, and shape factor based conduit conductance in comparison with the full DNS results on the entire images is shown in Fig. 12. The formation factor was calculated from the effective diffusivity of the domain. Applying a Dirichlet boundary condition for inlet and outlet concentration on each domain, effective diffusivity and formation factor were calculated as follows:

$$D_{eff} = \frac{J_A \times L}{\Delta C_n \times A},\ F = \frac{1}{D_{eff}}, \tag{10}$$

where J_A and F are the molar rate of species transport through the domain, and the formation factor of the domain, respectively.

The DL-based PNM can estimate the diffusive mass transport behavior in these porous media samples within acceptable accuracy while requiring considerably less computational time in contrast to the DNS method. After extracting the images of the conduits, the average runtime to estimate the conduit conductance of 6542 image samples of 3D conduits using the DNS method was 3.5 h, whereas the predictions using the trained DL model was done in 16.5s.

R^2 accuracy of each PNM approach to estimate formation factor in contrast to DNS on the entire medium for these 15 cases were: 1.17, 0.49, 0.87, and 0.84 for geometry-based conductance model, shape factor based model, DNS-based model, and DL-based model, respectively. Assuming a baseline of 0 for R^2 accuracy of a model that always predicts the average value of a target, a negative R^2 can be analyzed as a scenario where the model predicts worse than the baseline.

The comparison shown in Fig. 12 demonstrates the advantage of DL-based PNM models both in terms of time and accuracy. However, the domain size for the 15 presented samples were not large enough to capture the significance of DL-based PNM in terms of time in contrast to the finite difference-based PNM approach. For this purpose, a separate comparison was applied on randomly generated porous samples of the same porosity (0.75) with different sizes. As the size of the samples increases, the advantage of runtime saving in DL-based PNM becomes

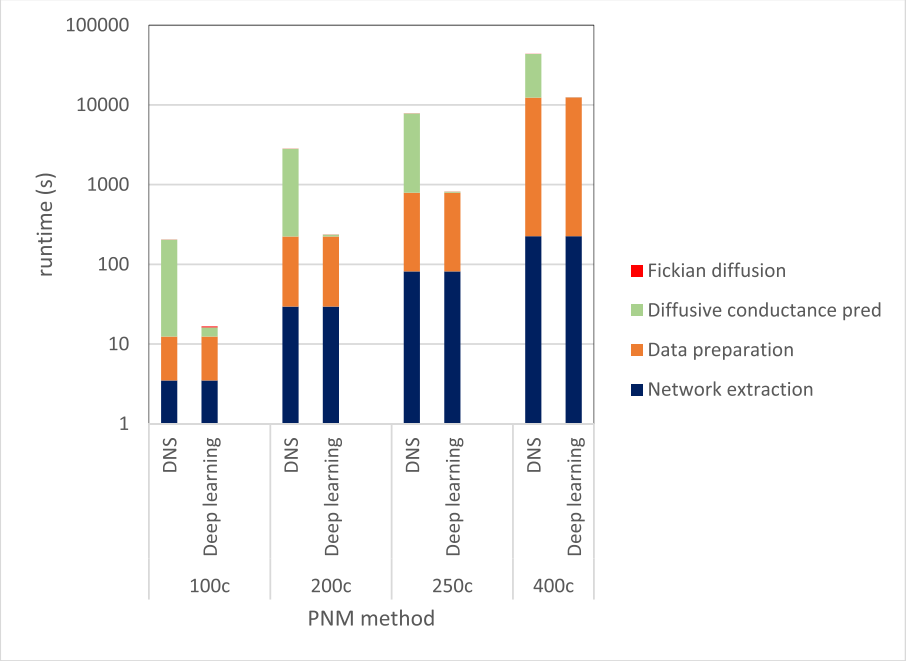


Fig. 13. Runtime to create a PNM and implement a Fickian diffusion transport through the domain using finite difference based conduit conductance method (DNS) and DL based method. Each bar indicates the steps applied on 4 randomly generated porous sample with size of 100^3, 200^3, 250^3, *and* 400^3. These steps include extracting the network, preparing images of conduits, predicting the diffusive conductance of conduits, and applying a diffusive mass transport throughout the network.

more significant as shown in Fig. 13. The runtime reported for PNM models includes pore network extraction, extraction of conduit pairs, prediction of conductance as well as Fickian diffusion simulation. The runtime for network and image extraction increases as the size of the domain becomes larger while the runtime for predicting diffusive conductance using DL-based model does not increase significantly. Despite the increase of time for extracting the network and conduit images, it should be noted that once the network is extracted, the same network can be used for fast simulation of transport phenomena and other simulation purposes in the porous sample.

4. Conclusion

A DL-based framework was developed to estimate diffusive conductance of conduit elements in pore network models using only images of pore-to-pore regions as the input data. The proposed framework is the first to estimate diffusion conductance rather than hydraulic conductance, and crucially uses only the images rather the more complex metrics obtained from images, greatly simplifying and accelerating its end use. The capability of the DL model to estimate the conduits' diffusive conductance was assessed with a non-biased test dataset. The test dataset contained a large number of porous media realizations with various morphologies, topologies, and geometrical complexities. The accuracy of the DL model to estimate diffusive conductance of irregular-shaped conduits was compared with the accuracy of conventional methods to calculate conductance values including geometry-based and shape factor based approaches. The DL model predicted diffusive conductance of test images within 0.94 R^2 accuracy, whereas geometry-based and shape factor based methods estimated conductance values within 0.62 and 0.71 R^2 accuracy, respectively. The speedup in predicting diffusive conductance using the DL model compared to computing them using DNS was on average 500x. Therefore, the developed framework was both massively faster than DNS and much more accurate than geometry-based methods. Ultimately, the predicted formation factor and/or effective diffusivity of full extracted networks was much more accurate than using geometry-based methods for conductance, leading to more predictive pore network models based on extractions from tomograms. The framework was trained on randomly generated sphere pack materials. For other types of porous material samples, the trained model can be retrained on a portion of new samples using transfer learning and can be used for prediction on the rest of the samples. Retraining the model would also offer a reduced training time as the current model's weights and trainable parameters need only be fine-tuned with new samples. The proposed DL framework was included in PoreSpy's pore network extraction module (porespy.network), and they are compatible with OpenPNM's diffusive conductance models for pore network modeling of transport phenomena. The dataset and codes of the proposed framework will be published online in the PMEAL group's github profile (PMEAL, n.d.).

Code availability section

PoreSpy's network extraction module extension Contact: jgostick@uwaterloo.ca.

Program language: Python

Software required: Tensorflow ("TensorFlow," n.d.), PoreSpy (Gostick et al., 2019).

The source codes will be available for downloading at the following link: https://github.com/PMEAL/porespy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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